



Inflatable Electroadhesive Clutches

Sadi Eren Erenel, Andrew Kim, Damian Gonzalez, and Prof. J. Edward Colgate

Northwestern

CENTER FOR
ROBOTICS AND
BIOSYSTEMS

Problem & Goal

Traditional clutches (mechanical, pneumatic, magnetic):

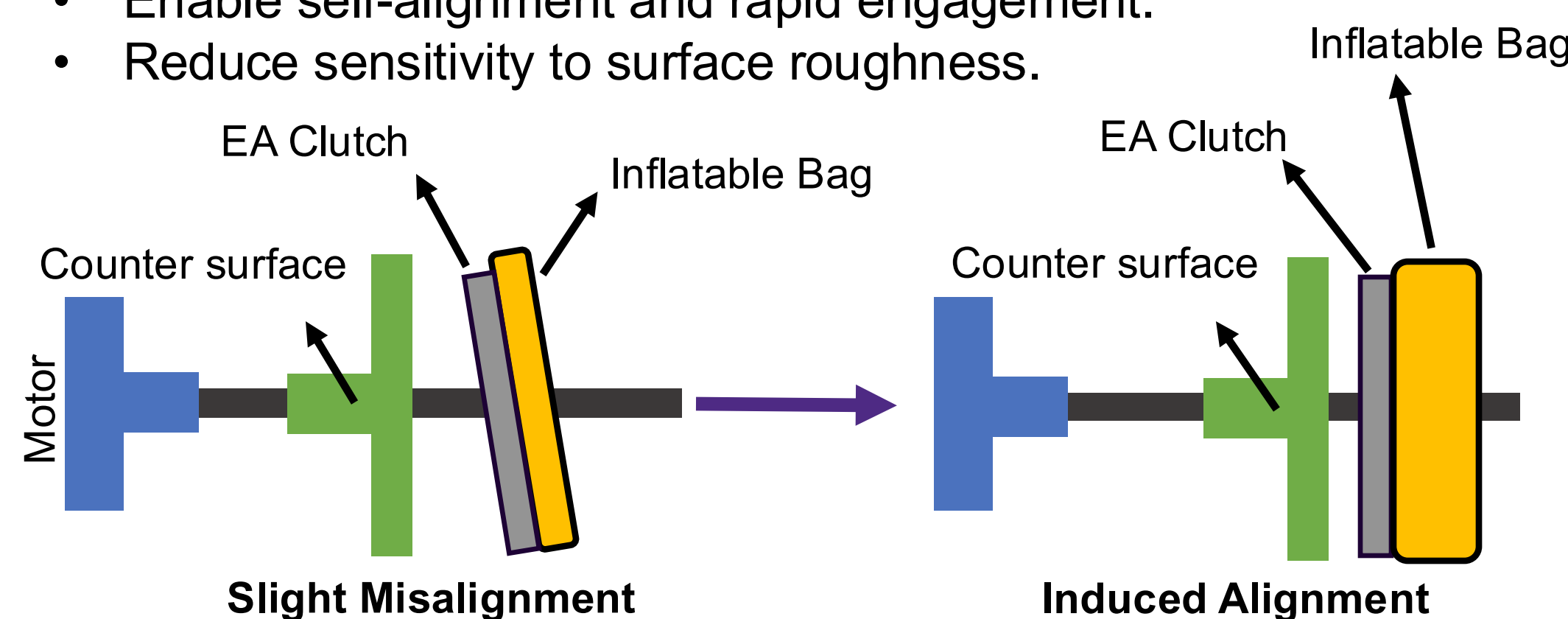
- Heavy and bulky — adds mass to mobile and wearable systems.
- Energy-inefficient — require constant actuation force or slip-based torque transfer, wasting power as heat.
- Slower to respond — limits performance in dynamic tasks.

EA clutches are lightweight, low-power, and fast — but:

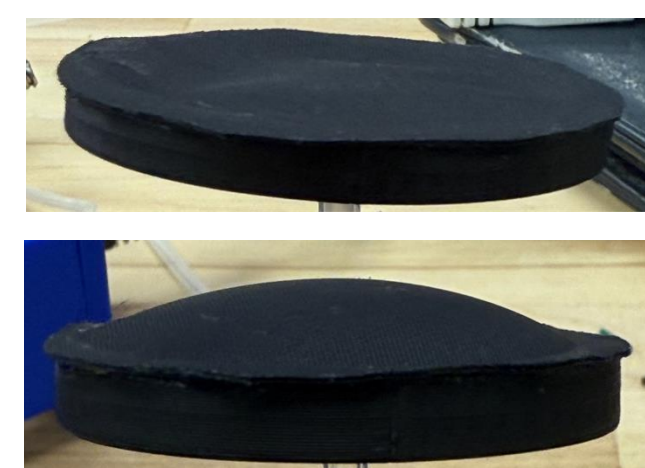
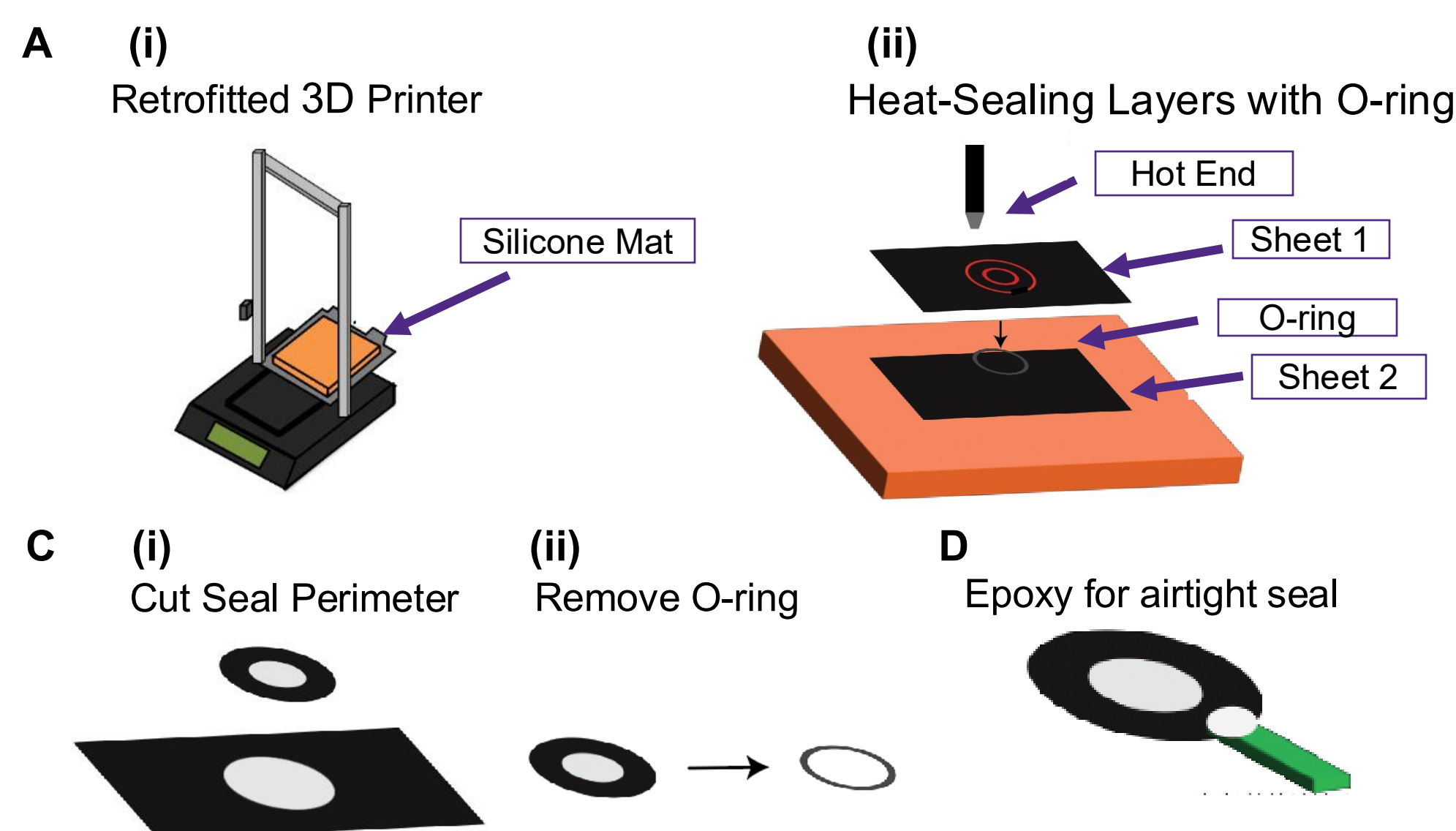
- Performance highly sensitive to surface smoothness & precise alignment.
- Misalignment causes reduced adhesion and holding force.
- **Non-uniform stress** leads to peeling and delamination over time.

Goal:

- Achieve **uniform load distribution via conformity**.
- Enable self-alignment and rapid engagement.
- Reduce sensitivity to surface roughness.

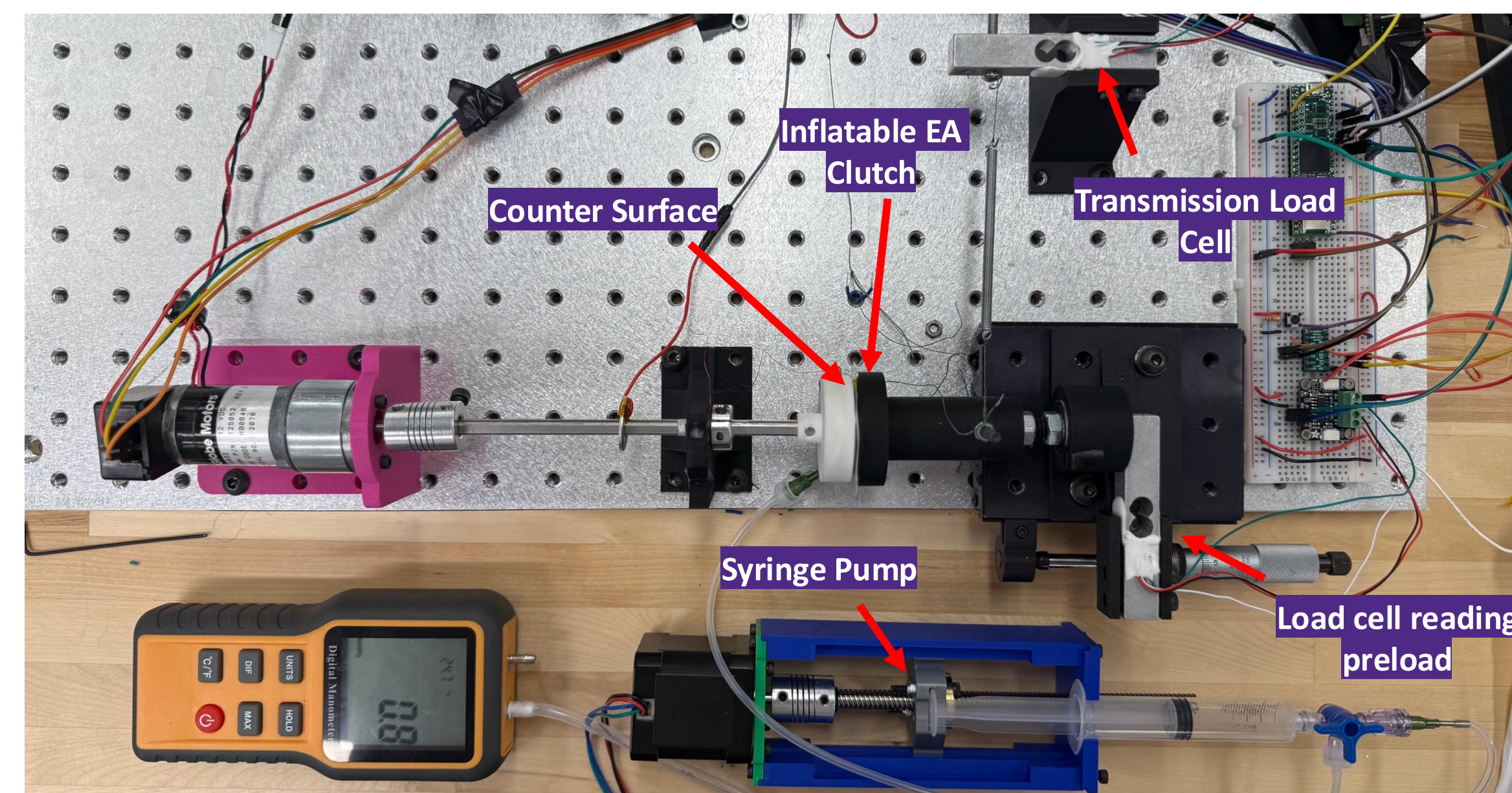


Inflatable Clutches Designs



Final Design for DAQ

Load Transmission Scheme



Goal: Quantify engagement/disengagement dynamics and transmit load with the inflatable clutch under motor-driven rotation.

Parameters:

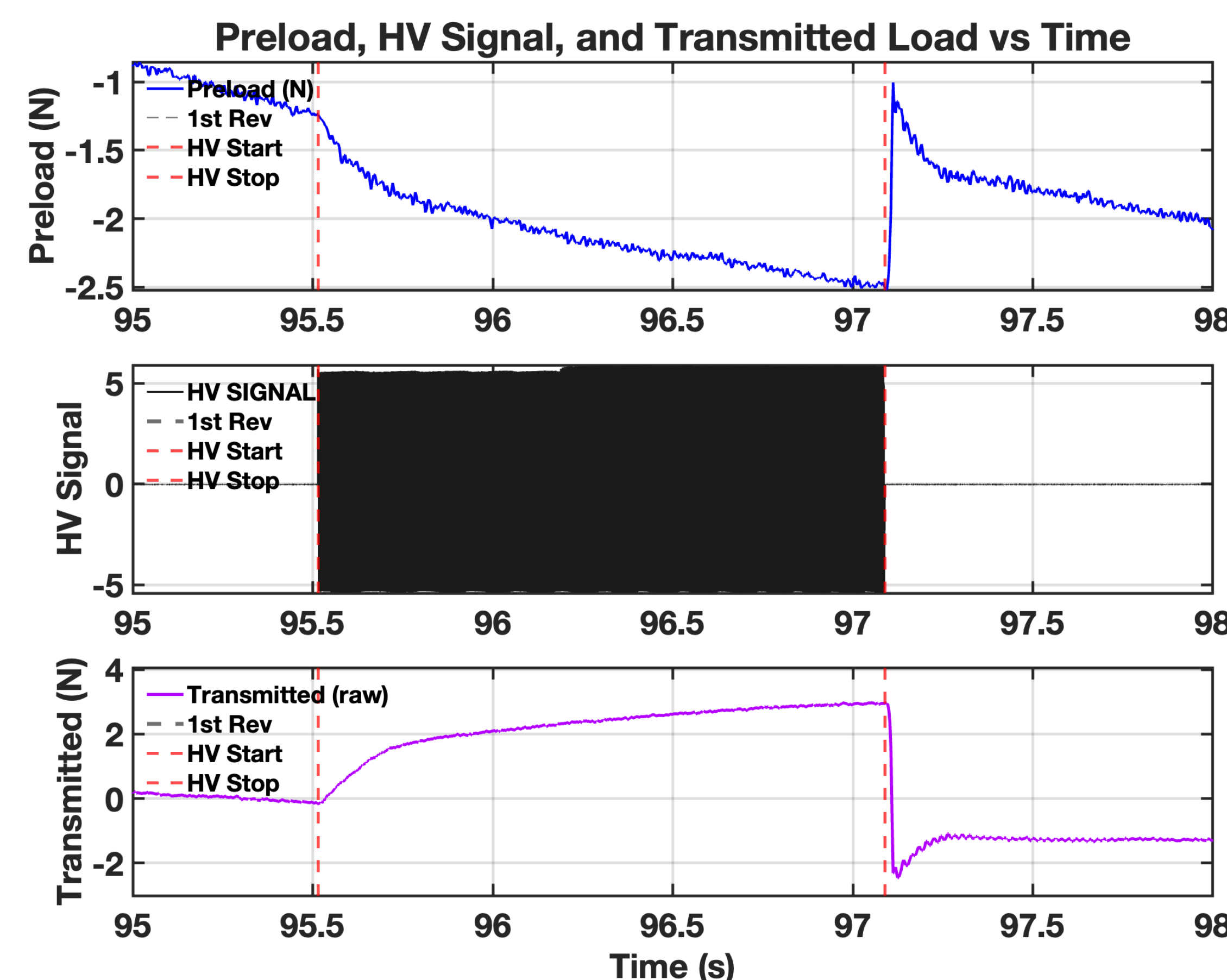
Used 50 mmHg inflation pressure
Gap Distance: 0.0762 mm

Protocol:

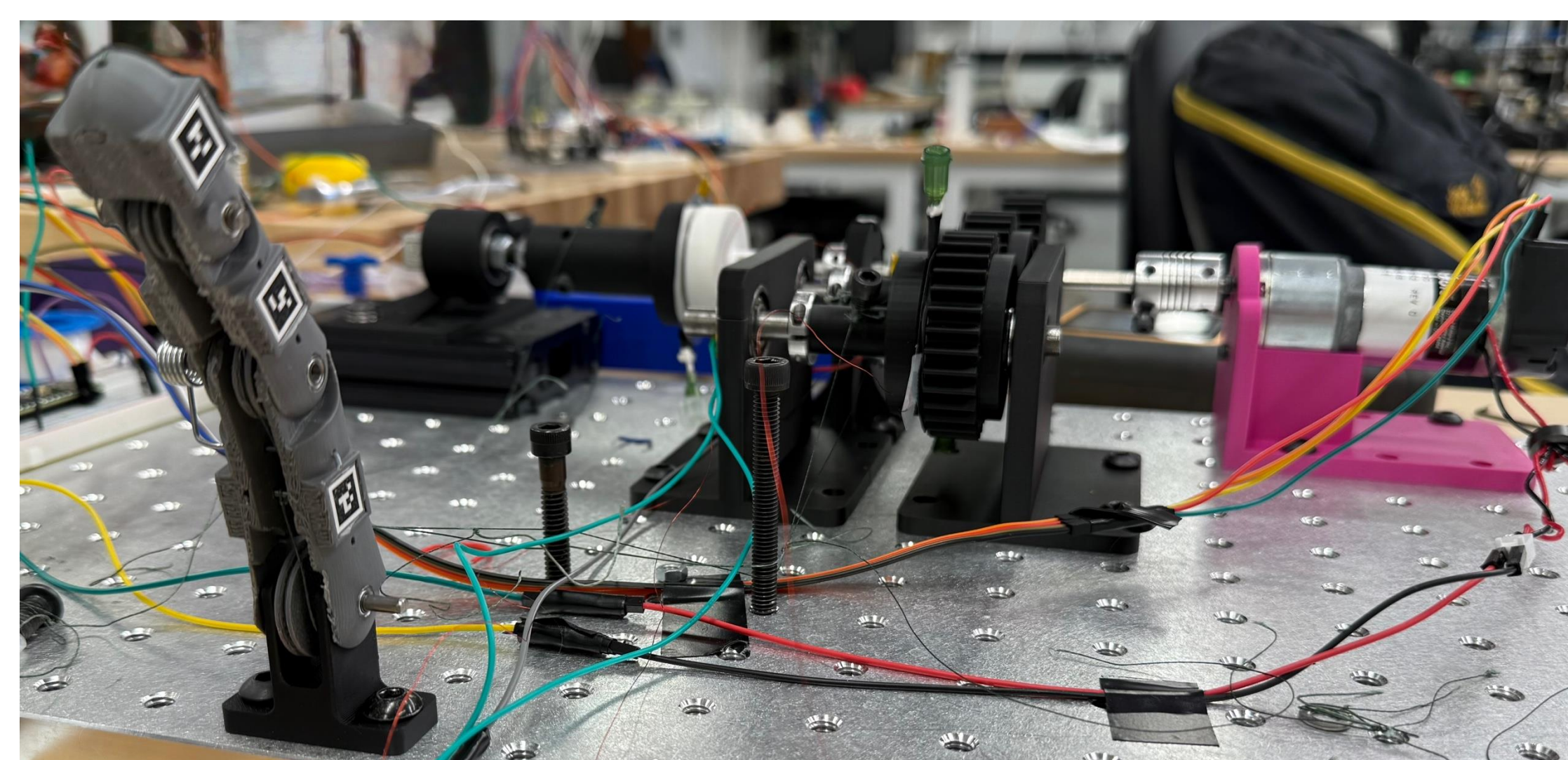
Measured engagement and disengagement times by recording the delay between HV activation/deactivation and the start/cessation of load transmission.

Results:

Engagement Time:
 0.657 ± 1.563 ms
Disengagement Time:
 8.360 ± 1.563 ms



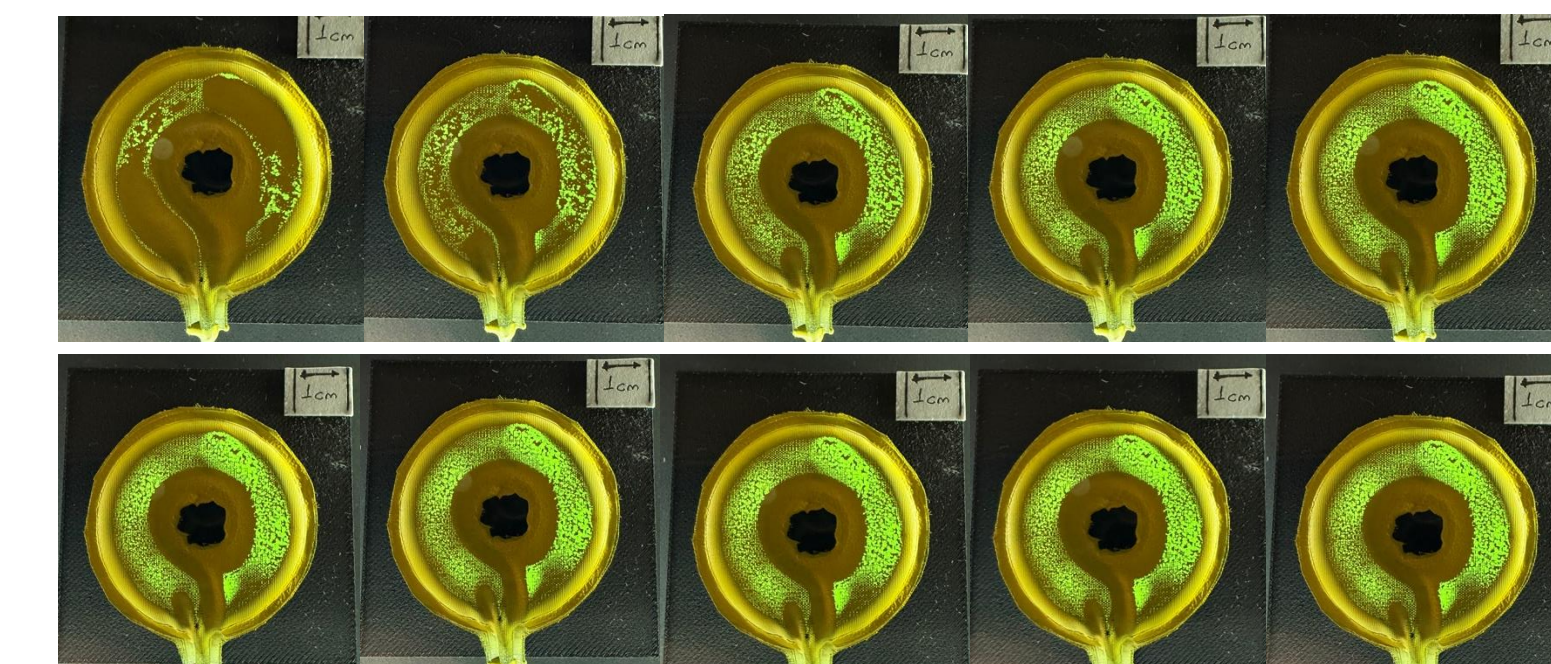
Application



- **3-DOF robotic finger** using multiple inflatable EA clutches
- **Independent inflation** and HV control for selective torque transfer and joint locking
- **Fast actuation** from preloaded EA engagement

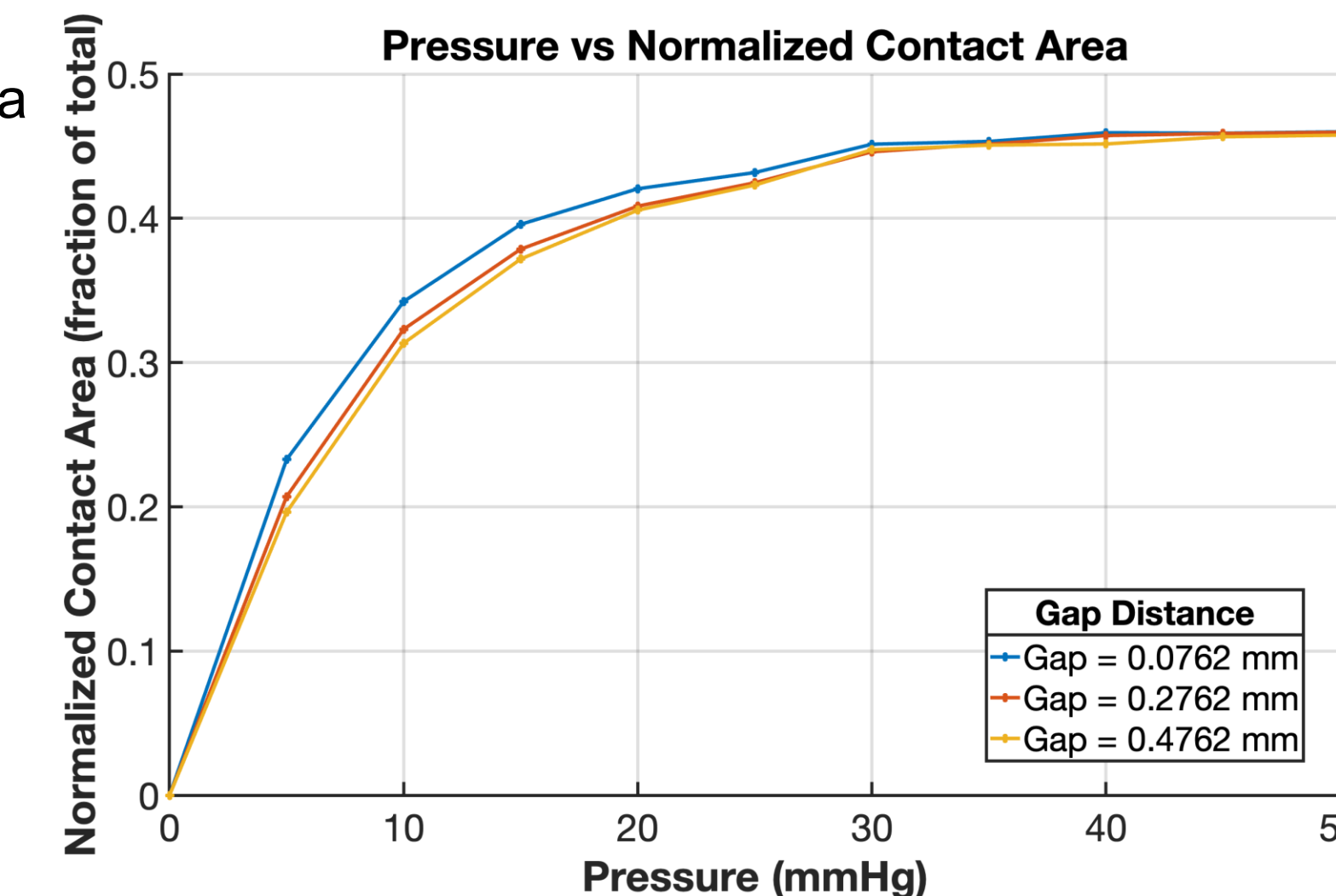
Contact Area

- **Method:** Used Frustrated Total Internal Reflection (FTIR) imaging to visualize EA clutch–substrate contact under varying inflation pressures (5–50 mmHg).

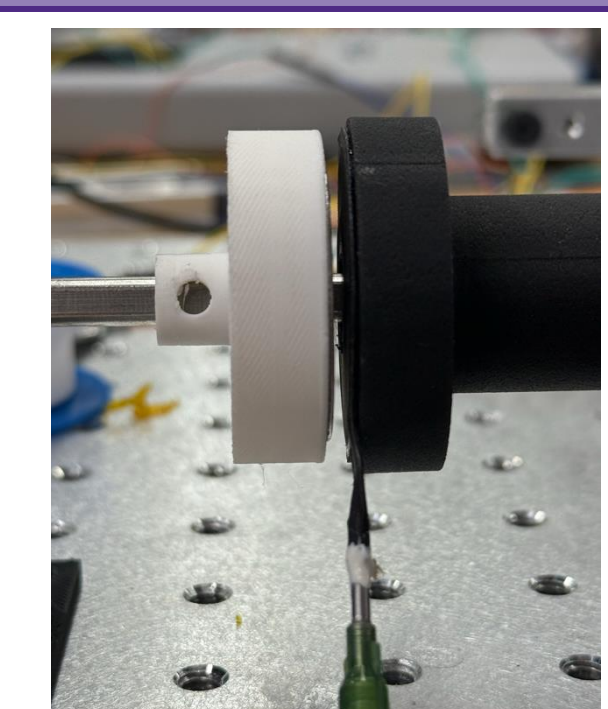
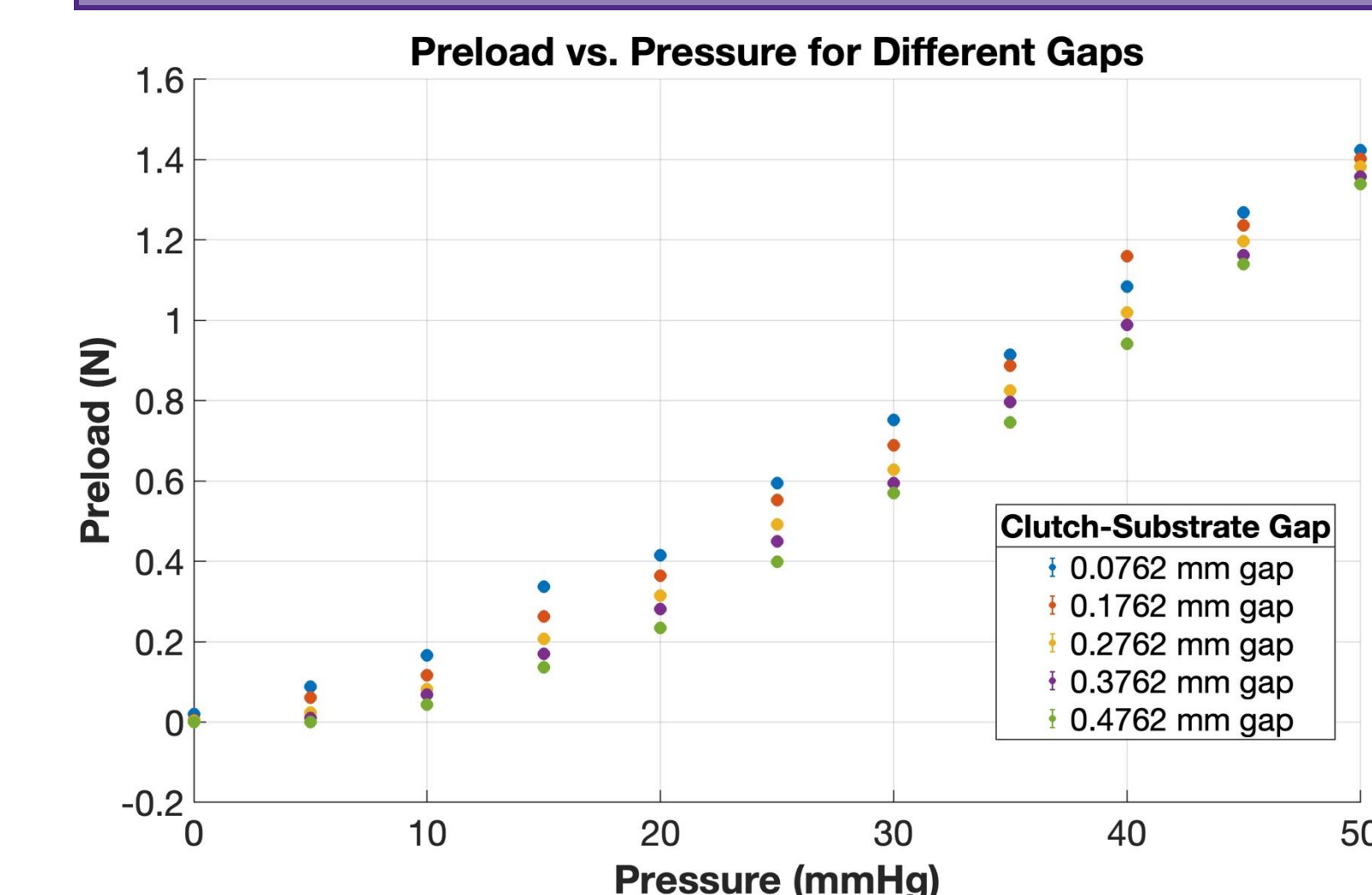


Coated Inflatable bags from 5 to 50 mmHg inflation

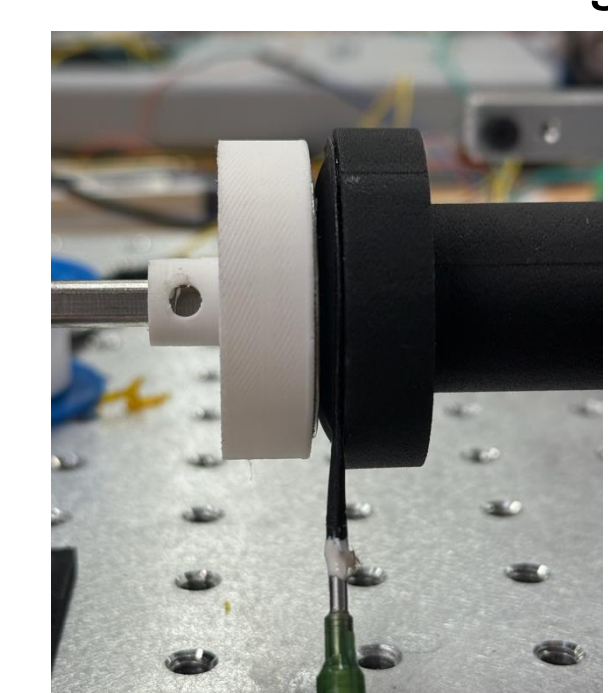
- **Finding:** Contact area plateaus at ~46% of the total surface around 30 mmHg, regardless of initial gap distance.
- **Implication:** Maximum achievable contact area is **gap-insensitive** once inflated enough.



Preload



Inflatable at 0 mmHg



Inflatable at 50 mmHg

- $\uparrow$ **Pressure** $\rightarrow$ $\uparrow$ **Preload**
- At any given pressure, larger gaps produce lower preload.
- Gap effect diminishes at high pressures as curves converge $\rightarrow$ **gap-insensitive regime**.

Future Work

- **Integrate bellow design** – Improve conformity and maintain uniform contact during actuation.
- **Max load characterization** – Extend pre-disengagement to clutch failure for true capacity.
- **Eliminate leakage** – Enhance sealing for stable inflation pressure.
- **Full finger control** – Coordinate clutch actuation for independent joint control in the 3-DOF finger prototype.

